

CSCE 633 MACHINE LEARNING · MODULE 01 · LECTURE NOTES

What Learning From Data Means

Optimising a proxy for a goal, and hoping it transfers.

MODULE OUTCOMES

- State the supervised learning problem formally.
- Explain the i.i.d. assumption and where it fails.
- Distinguish the loss you optimise from the goal you have.
- Explain why a held-out set is the only honest measurement.
- Decide whether a problem needs machine learning at all.

1 The setup

$$f^* = \operatorname{argmin}_f E_{(x,y) \sim D} [L(f(x), y)] \quad \text{but you compute} \quad (1/n) \sum_i L(f(x_i), y_i)$$

You are given a sample of n pairs drawn independently from an unknown distribution D . **You want the function minimising expected loss over D — the risk. You can only compute the average loss over the sample — the empirical risk. Minimising the second in the hope of minimising the first is the entire activity**, and the distance between them is what Modules 03 and 04 are about. It is worth sitting with how strange this is compared with everything else in this program: there is no specification to satisfy and no proof of correctness available, only an estimate of how well you will do on data you have not seen.

THREE SUBSTITUTIONS, EACH OF WHICH CAN FAIL

The sample stands in for the distribution. This fails whenever the data was collected in a way that does not match how the model will be used — which is the usual case, because data is collected by whatever process was convenient and models are deployed wherever they are useful.

The loss function stands in for the goal. Squared error is not 'good predictions'; cross-entropy is not 'useful'; accuracy is not 'helpful to the user'. **You optimise what you can write down**, and the gap between that and what you wanted is permanent (§3).

Past data stands in for future data. This fails whenever the world changes — and **especially when the model's own deployment changes it**, which is Module 13's feedback loop and the most insidious of the three.

Every machine learning failure is one of these three, and naming which one is most of a diagnosis. A model that worked in development and fails in production has almost always violated the first or the third, and knowing which determines whether the fix is better data or more frequent retraining.

2 The i.i.d. assumption

Violation	Example	Consequence
Temporal correlation	Stock prices, sensor streams, user activity logs — adjacent rows are nearly identical.	A random split puts the future in the training set , so the model is evaluated on data it effectively already saw. Split forward in time, always.
Grouped data	Several scans per patient; several sessions per user; several frames per video.	Split by group, or the same entity appears in train and test and the score measures memorisation.
Selection bias	Only approved loans have repayment outcomes. Only treated patients have treatment results.	The model never sees the rejected cases , so it cannot learn about them — and it will be applied to them.
Covariate shift	Trained on one hospital's scanner, deployed on another's.	$P(x)$ changes while $P(y x)$ does not. Sometimes correctable by reweighting.
Label shift	Disease prevalence changes between seasons.	The base rates move , so a calibrated model becomes miscalibrated even though nothing about the features changed.
Feedback	The model's predictions change the data it is later trained on. A recommender shapes what gets watched.	Module 13, and the hardest case — the distribution is no longer exogenous.

THE RANDOM SPLIT IS THE WRONG DEFAULT MORE OFTEN THAN NOT

A random split assumes every row is independent of every other. When rows share a patient, a user, a session, a device, or a time window, they are not — and this is extremely common.

So the same entity appears in both the training and the test set, and the model is rewarded for recognising that entity rather than for generalising to new ones. The reported score measures something real; it just is not the thing you will deploy.

Split by the unit you intend to generalise over. If the model will see new patients, split by patient. If it will see the future, split by time and never shuffle. **Ask 'what will be new at prediction time?' and make that the split key.**

This single error inflates reported performance more than any other in the field. It is easy to make (the library default is a random split), hard to notice (the numbers look plausible, just too good), and **published results have been retracted over it** — including in medical imaging, where the consequences were real.

3 The loss and the goal

YOU OPTIMISE THE LOSS; YOU ARE JUDGED ON THE GOAL

'Predict well' is not a mathematical object and cannot be optimised. Squared error, cross-entropy, hinge loss, and absolute error are mathematical objects, so one of them is chosen to stand in for the goal.

And the choice has real consequences. Squared error grows quadratically, so it is dominated by the largest errors and the fitted model chases outliers; absolute error does not, and is less convenient to optimise. **The usual choice is made for differentiability and then rationalised.**

Asymmetric costs are the common case and the symmetric loss is the default. A missed fraudulent transaction and a falsely declined legitimate one cost very different amounts to very different people, and a symmetric loss asserts they are equal. **So does accuracy**, which is why Module 05 spends its time on other metrics.

So write down the cost of each kind of error before choosing a loss. If the business cannot state those costs, that is worth discovering in week one rather than after deployment — and the conversation frequently reveals that the stated goal was not the real one.

4 Whether to use machine learning at all

- **A rule works.** If three if statements reach 95% of the achievable performance, ship them — they are debuggable, explainable, testable, and they do not drift. **Google's first rule of machine learning is 'don't be afraid to launch a product without machine learning'**, and several of the next few are about heuristics.
- **You have no labels**, and acquiring them costs more than the problem is worth. Labelling is usually the expensive part and is usually underestimated.
- **The data is small.** Below a few hundred examples a flexible model mostly fits noise, and the variance of your *evaluation* is large enough that you cannot tell whether it worked (Module 05).
- **You cannot tolerate being wrong in ways you cannot predict.** A learned model's failure modes are discovered empirically, not derived — which is a poor fit for systems with hard safety requirements.
- **Or the cost of being wrong falls on someone who did not choose to be subject to it.** That is a reason for considerably more care in evaluation and monitoring (Module 13), and it is not automatically a reason to refuse — but it changes what an acceptable error rate is, and who gets to decide.

THE BASELINE IS NOT OPTIONAL

Always compute a trivial baseline before training anything: predict the majority class, predict the mean, predict last week's value, or use the single most obvious feature.

A surprising number of reported models do not beat one, and the paper or the dashboard rarely says so — because nobody computed it, which is a different and more forgivable failure than hiding it.

A baseline also calibrates the problem. If the majority class is 94% of the data, then a model reporting 95% accuracy has done one percentage point of actual work, and 'our model is 95% accurate' is a deeply misleading sentence that is nonetheless true.

So the baseline is computed first, reported beside the result, and is the number every claim is relative to. Every course in this program has had this rule in some form — CSCE 735 Module 07 insisted on an *optimised* serial baseline before claiming a speedup, for exactly the same reason. **A number without a comparison is not a result.**

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Stanford CS229 — lecture 1 and the supervised learning notes (free)	The §1 formulation, derived rather than asserted.
James et al. — An Introduction to Statistical Learning, chapter 2 (free PDF)	The best free statement of §1 and §3 , with the accuracy-versus-interpretability trade laid out honestly.
Google — Rules of Machine Learning (free)	The §4 material. Rules 1 through 3 are about not using machine learning yet, which is the most useful advice in the document.
Kapoor & Narayanan — Leakage and the Reproducibility Crisis in ML-based Science (free)	The §2 callout, with a survey of how often it happens — across seventeen fields, and the answer is 'constantly'.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 State a problem you care about in the §1 formulation. Name D , L , and what the sample is.
- 2 **For that problem, name which of the three substitutions is weakest** and why.
- 3 Take a time series and evaluate a model with a random split and with a forward split. **Report both numbers and explain the gap.**

- 4 **Construct a grouped dataset** (several rows per entity) and show how much a random split inflates the score.
- 5 Fit a model with squared error and with absolute error on data containing outliers. Compare the fits visually.
- 6 **Write down the cost of a false positive and a false negative** for a problem you know, and construct a loss that reflects it.
- 7 Take a problem and **solve it with three if statements**. Report how far that gets you.
- 8 Compute majority-class and mean baselines for three datasets and report them.
- 9 **Find a published claim of model accuracy** and determine whether the baseline was reported.
- 10 Write your project's evaluation protocol — split, metric, baseline — and commit it before training anything.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 State the supervised learning problem and say which quantity you can actually compute.
- 2 Name the three substitutions and give a failure of each.
- 3 Give six ways the i.i.d. assumption fails.
- 4 Why is a random split usually wrong, and what should you split by?
- 5 Why is the loss not the goal, and what should you do about it?
- 6 Give five reasons not to use machine learning.
- 7 Why is a trivial baseline mandatory, and what does it calibrate?